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EXPERIMENT: Advanced SQL Queries

AIM: To enhance SQL problem-solving abilities by solving practical database questions using joins, grouping, filtering, subqueries, and aggregate functions.

Objective:
- Understand practical SQL problem-solving techniques.
- Apply JOIN, GROUP BY, HAVING, and aggregate functions.
- Retrieve missing records using LEFT JOIN.
- Use filtering and sorting to generate analytical reports.
- Develop optimized SQL queries for real-world scenarios.

Question 1 (Easy):
Matching Skills Find candidates who possess all the required skills: Python, Tableau, and PostgreSQL.

Solution:
```
SELECT candidate_id FROM candidates
WHERE skill IN ('Python','Tableau','PostgreSQL')
GROUP BY candidate_id HAVING COUNT(DISTINCT skill)=3
ORDER BY candidate_id;
```

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Question 2 (Medium):
Pages With No Likes Find the IDs of Facebook pages that have never received a like.

Solution:

```
 SELECT p.page_id FROM pages p
LEFT JOIN page_likes pl
ON p.page_id=pl.page_id
WHERE pl.page_id IS NULL
ORDER BY p.page_id;
```

Question 3 (Hard):
Tesla Unfinished Parts Find the part names and the number of unfinished parts whose finish date is NULL.

Solution:

```
 SELECT part, COUNT(*) AS unfinished_parts
FROM parts_assembly
WHERE finish_date IS NULL
GROUP BY part
ORDER BY unfinished_parts DESC, part;
```

Learning Outcomes:
- Write SQL queries using joins and aggregate functions.
- Apply GROUP BY and HAVING for grouped analysis.
- Retrieve unmatched records using LEFT JOIN.
- Solve real-world SQL interview problems efficiently.
- Improve database query writing and optimization skills.